Statistics 651
Posterior Inference

Winter 2026

BYU
Stats

Objectives

1. Justify Bayesian point estimates with standard loss functions
2. Summarize posterior distributions from analytic and Monte Carlo perspectives
3. Find regions of highest posterior density (HPD)
4. Conduct hypothesis tests
5. Propagate uncertainty from the posterior to related quantities of interest
6. Test models through simulation

Resources

- BDA3:
  - Ch 2.3 (posterior summary)
  - Ch 3.7 (bioassay experiment)
  - Ch 4.4 (frequency evaluation of Bayesian methods)
  - Ch 7.4 (Bayes factors)

Joint and marginal posterior distributions

Full joint posterior distributions can be difficult to summarize.

We often view inferences via marginal posterior distributions.

Definition

Given a joint posterior distribution $\Pi(\boldsymbol{\theta} \mid \boldsymbol{x})$ for $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)$, the **marginal posterior distribution** of θ_k is

$$\Pi_k(\theta_k \mid \boldsymbol{x}) = \int_{\boldsymbol{\Theta}_{-k}} \Pi(\theta_k, \mathrm{d}\boldsymbol{\theta}_{-k} \mid \boldsymbol{x}) \,,$$

for $k = 1, \ldots, K$, where $\boldsymbol{\theta}_{-k} := (\theta_1, \ldots, \theta_{k-1}, \theta_{k+1}, \ldots, \theta_K)$.

Decision theory, revisited

Decision theory in estimation

Recall the loss function

$$L(\theta, a) > -\infty$$

where

- $\theta \in \Theta$ is the state of nature, and
- $a \in A$ is an action that can be taken.

In statistical (point) estimation problems, $a \in \Theta$.

Decision theory in estimation

Definition

Common loss functions in point estimation

Squared Error Loss

$$L(\theta, a) = (\theta - a)^2$$

Absolute Error Loss

$$L(\theta, a) = |\theta - a|$$

0-1 Loss

$$L(\theta, a) = 1\{\theta \neq a\}$$

Expected loss revisited

Definition

Given a loss function $L(\theta, a) \in \mathbb{R}$ and probability distribution Π on θ (e.g., prior or posterior), the **expected loss** for any action is

$$\mathbb{E}_{\Pi}[L(\theta, a)] := \int_{\Theta} L(\theta, a)\, \Pi(\mathrm{d}\theta)\,.$$

A **Bayes estimator**, $\hat{\theta}$, minimizes posterior expected loss

$$\hat{\theta} = \mathrm{argmin}_a\, \mathbb{E}_{\Pi(\theta|x)}[L(\theta, a)]\,.$$

Bayes estimators

Examples

Find the Bayes estimator for each of the common loss functions below

Squared Error Loss

$L(\theta, a) = (\theta - a)^2 \longrightarrow$ Posterior mean $E(\theta \mid \underset{\sim}{x})$

Absolute Error Loss

$L(\theta, a) = |\theta - a| \longrightarrow$ Post median

0-1 Loss

$L(\theta, a) = 1\{\theta \neq a\} \longrightarrow$ Maximum a posteriori

$\rightarrow$ Maximize posterior dens/pmf

Post mean minimizes squared error loss

$$E_\theta\left[(\theta - a)^2\right] = E\left[\theta^2 - 2a\theta + a^2\right]$$

$$= \underbrace{E(\theta^2)}_{\text{expand}} - 2aE(\theta) + a^2$$

$$= Var(\theta) + \underbrace{E(\theta)^2 - 2aE(\theta) + a^2}$$

$$= Var(\theta) + \left(E(\theta) - a\right)^2$$

$a = E(\theta)$ minimizes this

Point estimation

Posterior mean

- Most common point estimator
- Easy to summarize with samples because $[\mathbb{E}(\boldsymbol{\theta} \mid \boldsymbol{x})]_k = \mathbb{E}(\theta_k \mid \boldsymbol{x})$, i.e., the kth element of the posterior mean is equal to the marginal posterior mean of θ_k
- $\mathbb{E}(h(\theta) \mid \boldsymbol{x}) \neq h(\mathbb{E}(\theta \mid \boldsymbol{x}))$ $\longrightarrow$ if I want $E(h(\theta) \mid \underline{x})$ w/ samples I need to apply $h()$ to each sample

Posterior median

- Denote as $Q_{\Pi(\theta|\boldsymbol{x})}(0.5)$
- $Q_{\Pi(h(\theta)|\boldsymbol{x})}(0.5) = h(Q_{\Pi(\theta|\boldsymbol{x})}(0.5))$
  if $h(\cdot)$ is continuous and monotone

Maximum a posteriori (MAP)

Definition

The **maximum a posteriori** (MAP) estimate of $\boldsymbol{\theta}$ is given as

$$\hat{\boldsymbol{\theta}}_{\mathrm{MAP}} = \mathrm{arg}\underset{\text{max}}{\cancel{\mathrm{min}}}_{\boldsymbol{\theta}}\, \pi(\boldsymbol{\theta} \mid \boldsymbol{x})\,.$$

Recall that $\pi(\boldsymbol{\theta} \mid \boldsymbol{x}) \propto \mathcal{L}(\boldsymbol{\theta}; \boldsymbol{x})\, \pi(\boldsymbol{\theta})$.

What is the MAP estimate under $\pi(\boldsymbol{\theta}) \propto 1$?

$$\hat{\theta}_{MAP} = \hat{\theta}_{MLE}$$

- The MAP estimate of $h(\theta)$ is not generally equal to $h(\hat{\theta}_{\mathrm{MAP}})$.

Frequentist evaluation of point estimators

Bayesians average loss functions with respect to a distribution on θ.

- $\mathbb{E}_\Pi[L(\theta, a)] := \int_\Theta L(\theta, a)\, \Pi(\mathrm{d}\theta)\,.$

Frequentists average loss functions with respect to a distribution on data $\boldsymbol{x}$.

- $\mathbb{E}_F[L(\theta, a)] := \int_{\mathcal{X}} L(\theta, a)\, F(\mathrm{d}\boldsymbol{x})\,.$

Any point estimator (Bayesian or otherwise) can be evaluated with $\mathbb{E}_F[L(\theta, a)]$, also called the **risk**.

Frequentist evaluation of point estimators

Example

Let $y \mid \theta \sim \text{Binomial}(n, \theta)$ and assume $\theta \sim \text{Beta}(\alpha, \beta)$.

$$= \frac{\alpha'}{\alpha' + \beta'} = \frac{\alpha + y}{\alpha + y + \beta + n - y}$$

Evaluate the risk for

- the posterior mean Bayes point estimator $\hat{\theta}_B$ and
  post mean
- the maximum-likelihood estimator $\hat{\theta}_{ML}$

$$\hat{\theta}_B = \frac{\alpha+\beta}{\alpha+\beta+n}\left(\frac{\alpha}{\alpha+\beta}\right) + \frac{n}{\alpha+\beta+n}\left(\frac{y}{n}\right)$$

$$= \frac{m}{m+n}(\mu) + \frac{n}{m+n}\left(\frac{y}{n}\right)$$

$$\hat{\theta}_{ML} = \frac{y}{n}$$

where

$m = \alpha + \beta$

$\mu = \frac{\alpha}{\alpha+\beta}$

under squared error loss. That is, consider the mean squared error $\mathbb{E}_F\left[(\theta - \hat{\theta})^2\right]$.

Compare the two estimators under different prior scenarios.

$$E_F\left[(\theta - \hat{\theta})^2\right] = E_F\left[\theta^2 - 2\theta\hat{\theta} + \hat{\theta}^2\right]$$

$$= \theta^2 - 2\theta E(\hat{\theta}) + Var(\hat{\theta}) + E(\hat{\theta})^2$$

$$= \underbrace{Var(\hat{\theta})}_{\text{variance}} + \underbrace{\left(\theta - E(\hat{\theta})\right)^2}_{\text{bias}^2}$$

Risk for MLE

$$\mathrm{Var}\left(\frac{y}{n}\right) + \left(\theta - E\left(\frac{y}{n}\right)\right)^2$$

$$= \frac{1}{n^2} n\theta(1-\theta) + \left(\theta - \frac{1}{n} n\theta\right)^2$$

$$= \frac{\theta(1-\theta)}{n}$$

Risk for Posterior Mean

$$\mathrm{Var}\left(\frac{m}{m+n}\mu + \frac{n}{m+n}\left(\frac{y}{n}\right)\right) + \left(\theta - E\left[\frac{m}{m+n}\mu + \frac{n}{m+n}\left(\frac{y}{n}\right)\right]\right)^2$$

$$= \frac{n^2}{(m+n)^2 n^2} n\theta(1-\theta) + \left(\theta - \left(\frac{m}{m+n}\mu + \frac{n}{m+n}\left(\frac{n\theta}{n}\right)\right)\right)^2$$

$$= \frac{n\theta(1-\theta)}{(m+n)^2} + \left[\theta - \left(\frac{m}{m+n}\mu + \frac{n\theta}{m+n}\right)\right]^2$$

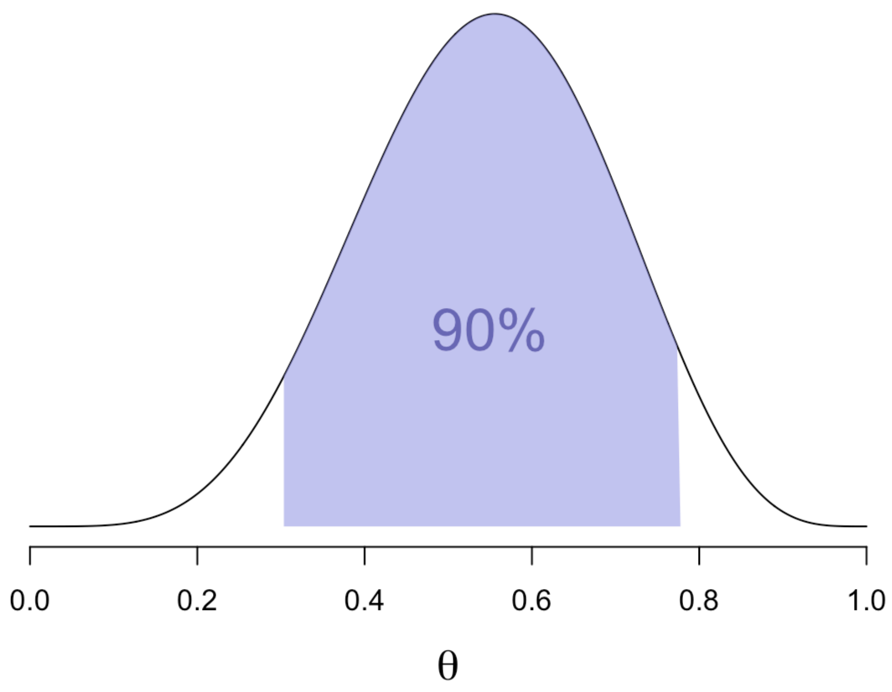
Credible regions

Credible regions (revisited)

Recall the definition of a posterior interval. This can apply to each θ_k marginally.

Definition

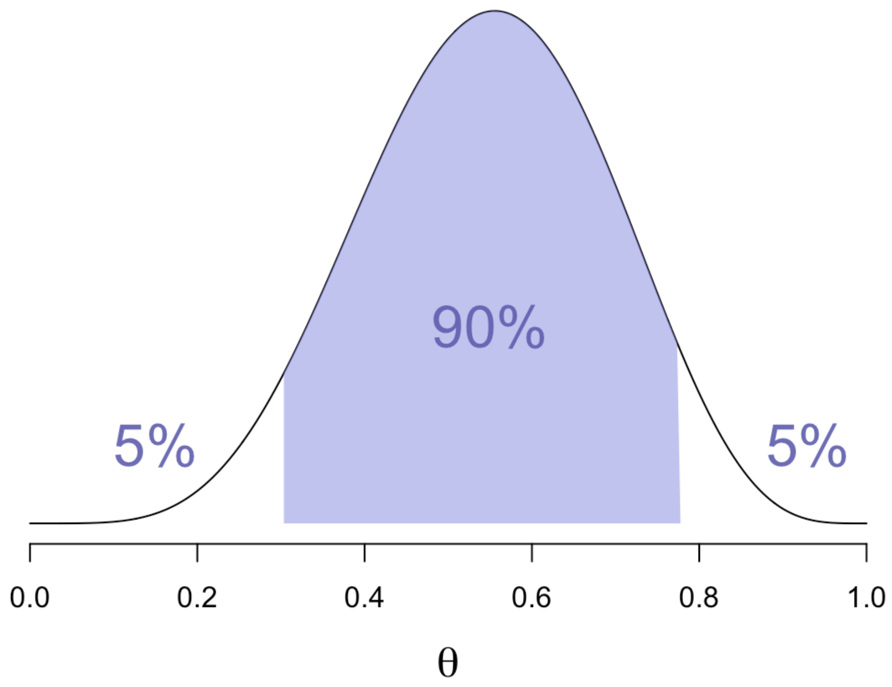
Any interval that covers $\gamma \times 100\%$ of the probability mass in the posterior distribution is called a $\gamma \times 100\%$ posterior (credible) interval.



Credible regions (revisited)

Definition

The $\gamma \times 100\%$ interval (a, b) for which $\mathbb{P}(\theta < a \mid \boldsymbol{x}) = \mathbb{P}(\theta > b \mid \boldsymbol{x}) = \frac{1-\gamma}{2}$ is called the $\gamma \times 100\%$ **equal-tail posterior interval.**



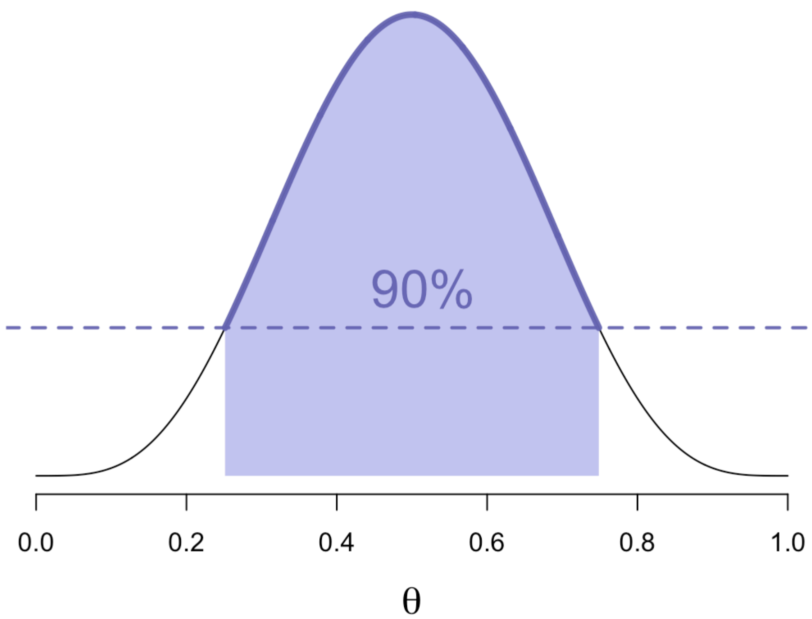
Highest posterior density (HPD) region

Definition

A region R for which

1. $\mathbb{P}(\boldsymbol{\theta} \in R \mid \boldsymbol{x}) = \gamma$, *and*
2. $\pi(\boldsymbol{\theta} \mid \boldsymbol{x}) > \pi(\boldsymbol{\theta}' \mid \boldsymbol{x})$ for all $\boldsymbol{\theta} \in R$ and $\boldsymbol{\theta}' \notin R$

is called a $\gamma \times 100\%$ **highest posterior density (HPD)** region.



Frequentist evaluation of interval estimators

The frequentist loss function for confidence regions is

$$L(\boldsymbol{\theta}, a) = 1\{\boldsymbol{\theta} \notin R\}\,,$$

where the action a is selecting confidence region R.

The risk function is then

$$\mathbb{E}_F(1\{\boldsymbol{\theta} \notin R)\})\,,$$

which is one minus the coverage probability under repeated sampling.

Any interval or region can be evaluated under this criterion.

Testing hypotheses

Posterior probability

Let H denote a hypothesis regarding $\boldsymbol{\theta}$ such as $\boldsymbol{\theta} \in R_H$.

The posterior distribution admits probability statements for H, i.e.,

$$\begin{aligned}\mathbb{P}(H \mid \boldsymbol{x}) &= \mathbb{P}(\boldsymbol{\theta} \in R_H \mid \boldsymbol{x}) \\ &= \int_{R_H} \Pi(\mathrm{d}\boldsymbol{\theta} \mid \boldsymbol{x})\,.\end{aligned}$$

Bayes factors

Consider the posterior probability of hypothesis H_1 (with $H_1^C = H_0$):

$$\mathbb{P}(H_1 \mid \text{data}) = \frac{\mathbb{P}(\text{data} \mid H_1)\, \mathbb{P}(H_1)}{\mathbb{P}(\text{data} \mid H_0)\, \mathbb{P}(H_0) + \mathbb{P}(\text{data} \mid H_1)\, \mathbb{P}(H_1)}$$

and write the posterior odds for H_1:

$$\text{Odds}(p) = \frac{p}{1-p}$$

$$\text{Odds for } H_1 = \frac{P(\text{data} \mid H_1)\, P(H_1)}{P(\text{data} \mid H_0)\, P(H_0)}$$

$$\frac{P(H_1 \mid \text{data})}{P(H_0 \mid \text{data})} = (BF_{H_1}) \frac{P(H_1)}{P(H_0)}$$

$$\Rightarrow BF_{H_1} = \frac{P(H_1 \mid \text{data}) / P(H_1)}{P(H_0 \mid \text{data}) / P(H_0)}$$

Bayes factors

Definition

The **Bayes factor** is given by

$$\frac{\mathbb{P}(\text{data} \mid H_1)}{\mathbb{P}(\text{data} \mid H_0)}$$

and represents the factor that takes us from prior odds $\mathbb{P}(H_1)/\mathbb{P}(H_0)$ to posterior odds.

Bayes factors

Kass and Raftery (1995) give an interpretation of the strength of evidence in favor of H_1 over H_0:

Bayes Factor	Strength of evidence
1 to 3.2	Not worth more than a bare mention
3.2 to 10	Substantial
10 to 100	Strong
above 100	Decisive

A similar (precursor) table appears in Harold Jeffreys' book Theory of Probability.

Derived quantities

Derived quantities

Interest often lies in functions of $\boldsymbol{\theta}$ (inference) or $\tilde{y}$ (prediction).

Propagating uncertainty in $\boldsymbol{\theta}$ to these **derived quantities** is conceptually straightforward with posterior distributions.

- Change of variables —> $p(h(\boldsymbol{\theta}) \mid \boldsymbol{x})$
- Simple with Monte Carlo samples

Derived quantities

Examples

In the bioassay example (BDA3 Ch 3.7)

- Logistic dose-response curve
- LD50, the dose at which the probability of death crosses 0.5

Examples

Posterior summary / inference

Bioassay example (BDA3 Ch 3.7)

Consider the toxicity data, where x_i is the dose and $y_i \in \{0, 1\}$ indicates survival (0) or death (1).

Fit the model $y_i \overset{\text{ind}}{\sim} \text{Binomial}(n_i, \theta_i)$ for $i = 1, \ldots, n$

with $\log\left(\frac{\theta_i}{1-\theta_i}\right) = \alpha + \beta x_i$.

a. Report posterior mean, median, standard deviation, and marginal 95% posterior intervals for each of α, β.

b. Find a joint 95% HPD region for (α, β). Compare this with the marginal intervals.

c. Find the posterior distribution of LD50, the dose at which the probability of death crosses 0.5.

d. Draw from the implied posterior distribution of dose-response curves.

e. Calculate the posterior probability and Bayes factor for the hypothesis $\beta > 0$: the drug is toxic.

Validating models via simulation

Linear regression example

Simulate fake data from the model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

with $\epsilon_i \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$ for $i = 1, \ldots, n$.

a. Fit the model with a PPL.

b. Check recovery of the parameters for different sample sizes, error variances.

c. Perform prior sensitivity analysis for various common priors for the error variance.

d. Repeatedly sample data sets from the model and evaluate the Bayes point estimators (in terms of mean squared error) and posterior interval coverage. Compare with the classical least squares estimates and confidence intervals.